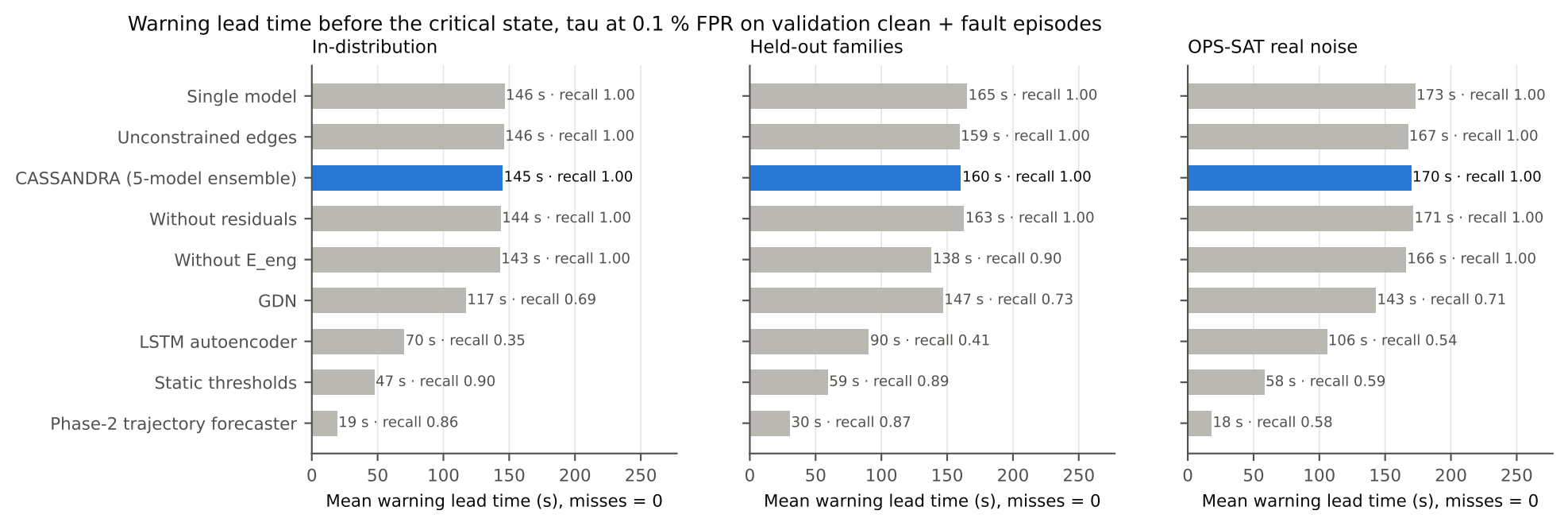
Warning lead time before the critical state, tau at 0.1 % FPR on validation clean + fault episodes
In-distribution
Held-out families
OPS-SAT real noise
Single model
Unconstrained edges
CASSANDRA (5-model ensemble)
Without residuals
Without E_eng
GDN
LSTM autoencoder
Static thresholds
Phase-2 trajectory forecaster
146 s · recall 1.00
146 s · recall 1.00
145 s · recall 1.00
144 s · recall 1.00
143 s · recall 1.00
117 s · recall 0.69
70 s · recall 0.35
47 s · recall 0.90
19 s · recall 0.86
165 s · recall 1.00
159 s · recall 1.00
160 s · recall 1.00
163 s · recall 1.00
138 s · recall 0.90
147 s · recall 0.73
90 s · recall 0.41
59 s · recall 0.89
30 s · recall 0.87
173 s · recall 1.00
167 s · recall 1.00
170 s · recall 1.00
171 s · recall 1.00
166 s · recall 1.00
143 s · recall 0.71
106 s · recall 0.54
58 s · recall 0.59
18 s · recall 0.58
0
50
100
150
200
250
Mean warning lead time (s), misses = 0